Question 1

In Szohôd, the capital of Borduria, a part of its 33 kV distribution network has a network problem that needs to be solved by the network operator/planner. The 4-bus network used for this analysis is illustrated in Figure 1. All lines have a reactance of 0.1 p.u. Line 1-2 has a thermal capacity of 45 MW; other lines have a thermal capacity of 32.5 MW. Lossless DC load-flow analysis could be performed to identify the problem and the solution options. The parameters of generators in the system are listed in Table 1 below.

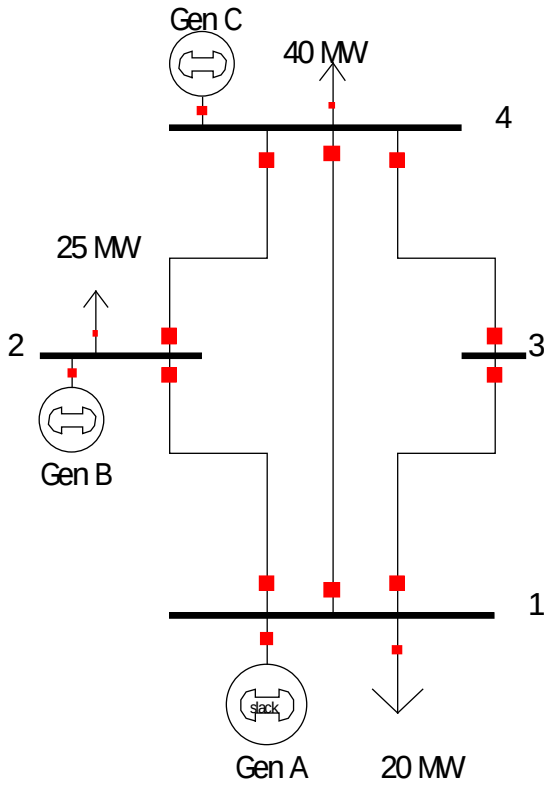


Figure 1: Szohôd distribution network

Table 1: Parameters of generators

Generator	Marginal cost (£/MWh)	Limit (MW)
Gen A	30	100
Gen B	50	20
Gen C	100	15

Based on the data above, please answer the following questions:

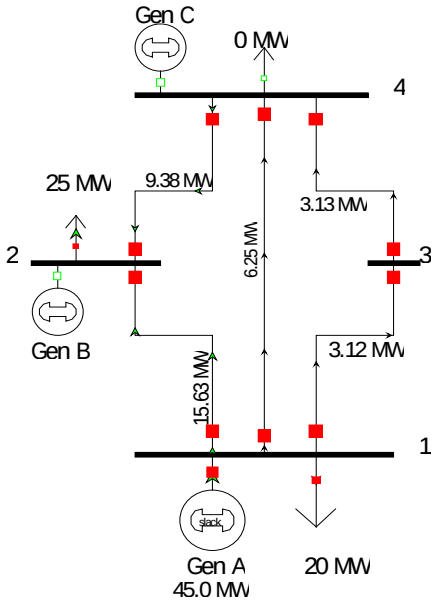
a) Determine the Economic Dispatch and compute the corresponding power flows in each of the circuits

[3]

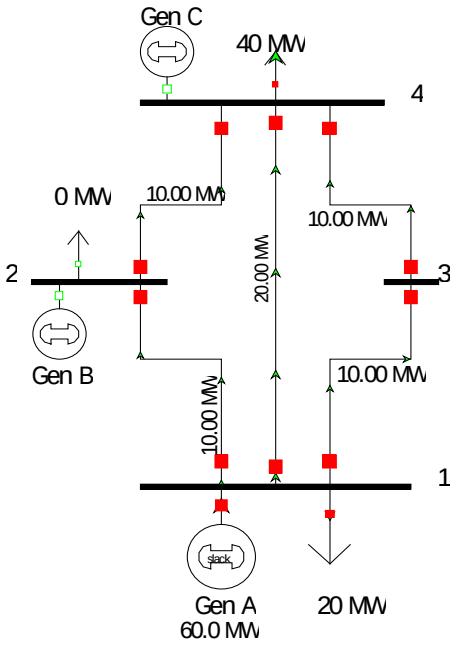
The Economic Dispatch solution: Gen A produces 85 MW. This costs 85 x 30 = £2,550/h

To obtain the power flows, we use the superposition approach:

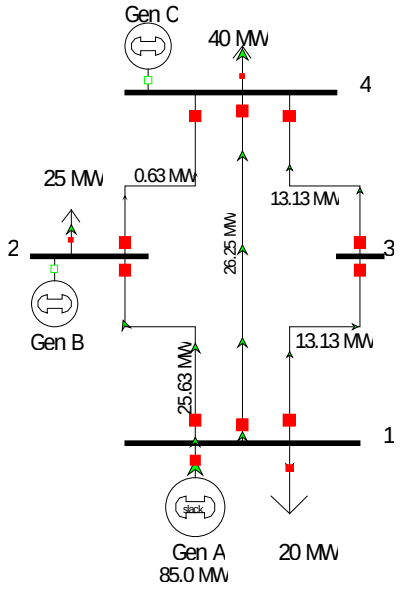
Step 1: Flows associated with load at bus 2



Step 2: Flows associated with load at bus 4



Imposing the flows obtained in step 1 and 2, the power flows in the system are obtained as shown in the diagram below.



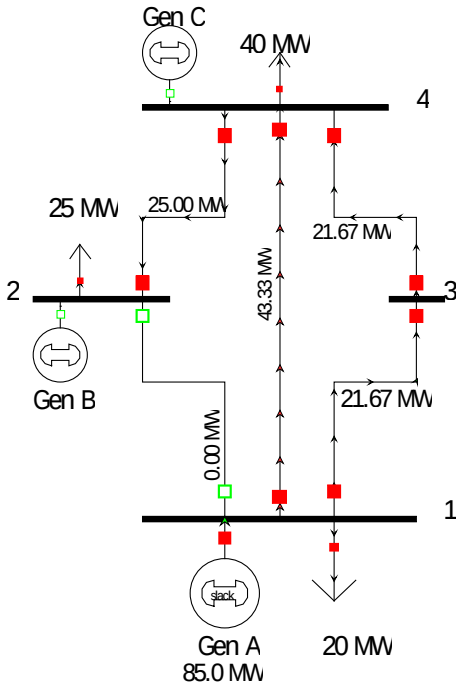
b) Determine the power flows following the outage of the individual circuit and identify conditions in which the power flows during the contingency exceed the circuit capacity limit.

[3]

Contingency analysis: (using the superposition technique to calculate the flows)

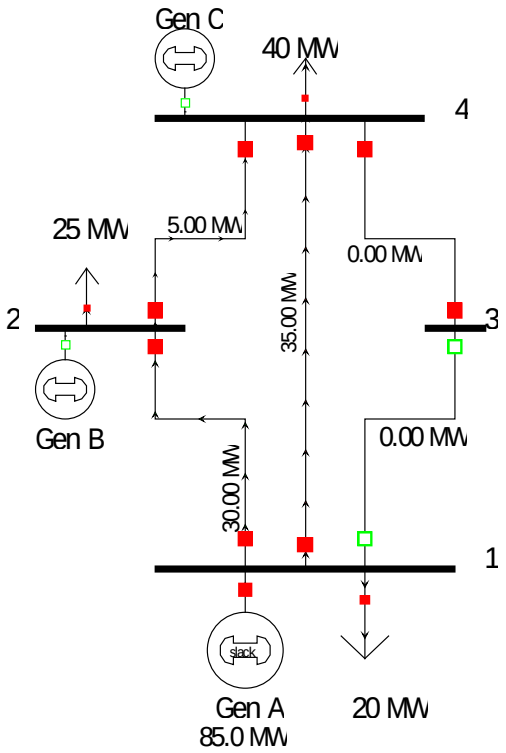
Line 1-2 outage

The outage leads into line 1-4 overload as shown in the diagram below.



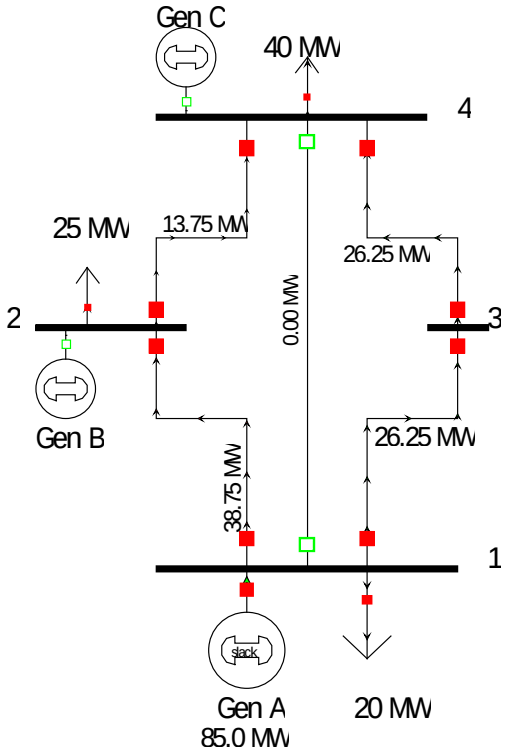
Line 1-3 outage (similar to line 3-4 outage).

Line 1-4 is overload.



Line 1-4 outage:

All intact lines are operated within their capacity limit.



c) If the system operator needs to secure the system in a preventive mode against one line outage using the out-of-merit generator, determine the least-cost secured dispatch and the corresponding cost.

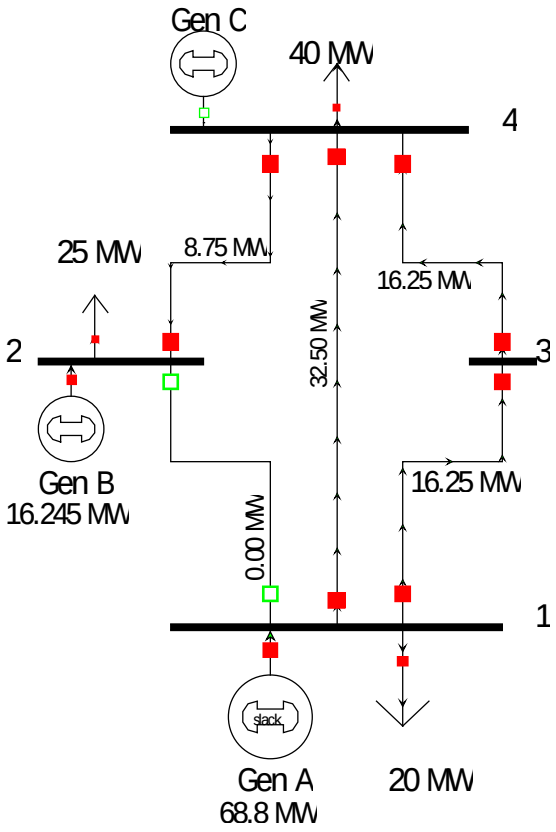
[6]

Based on the results of the contingency analysis, the most severe contingency is when Line 1-2 fails which leads to overload at Line 1-4. The sensitivity of the flows at line 1-4 with the injection at bus 1 is 2/3.

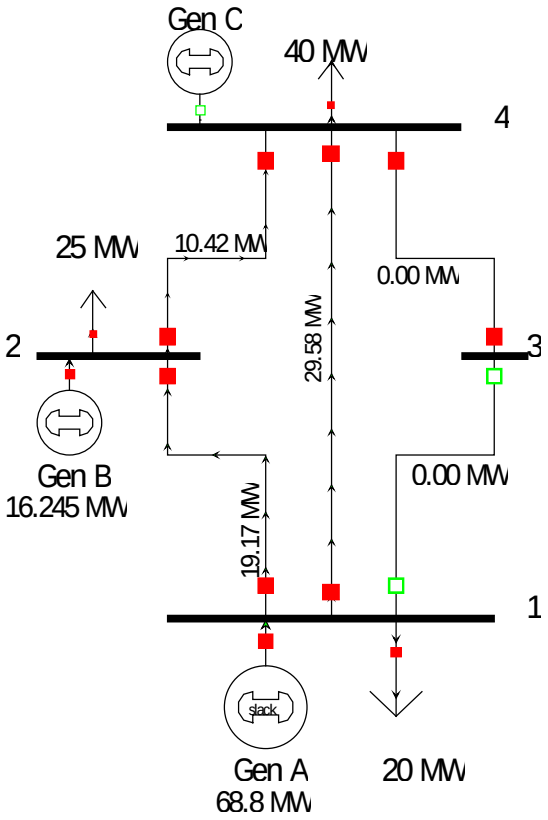
In order to reduce the flows back to the limit (32.5 MW), gen B needs to produce (43.33-32.5)*3/2 = 16.245 MW

Due to the redispatch, the cost increases from £2,550/h (Economic Dispatch) to { (85-16.245) x 30 + (16.245 x 50)} = £2874.90/h . The cost of security is 2874.90-2550 = £324.9/h

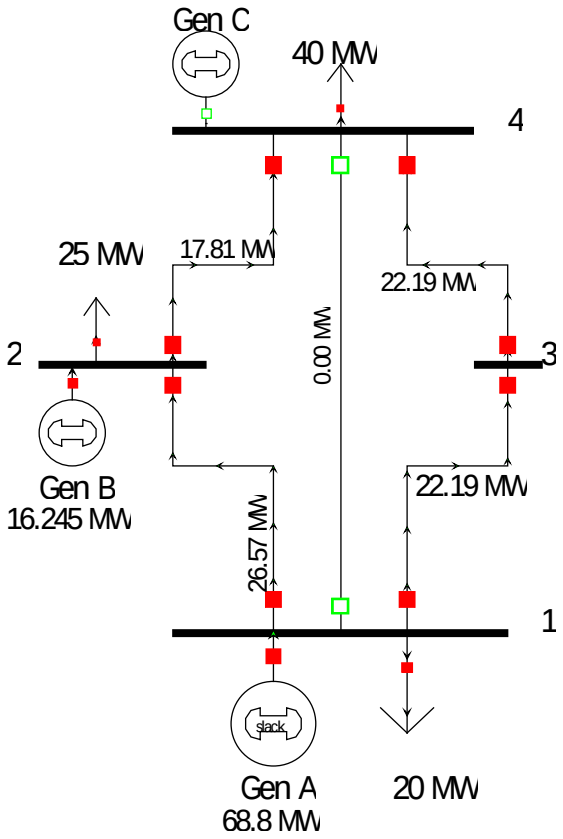
The impact of re-dispatching generators, the flows when line 1-2 outage are shown below.



This solution is analysed against the outage at Line 1-3 outage. The power flows are shown in the diagram below. It shows that the re-dispatch also solves the problem due to Line 1-3 outage.



This solution is analysed against the outage at Line 1-4 outage. The power flows are shown in the diagram below. All intact lines are operated within the limit.



d) Another option is to install a Quadrature-booster (QB) in the system to reduce its operating cost. Determine the line where the QB should be installed and the setting of the QB, so it meets the N-1 security criterion.

[8]

The critical contingencies are:

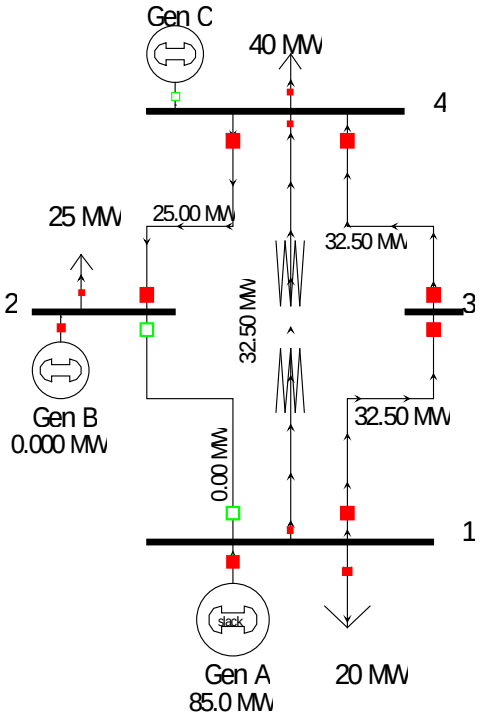
- Line 1-2 outage
- Line 1-3 outage

And the line affected by those contingencies is Line 1-4. Given that there is still a spare capacity at line 1-2 and line 1-3; the best place to install a QB is at line 1-4.

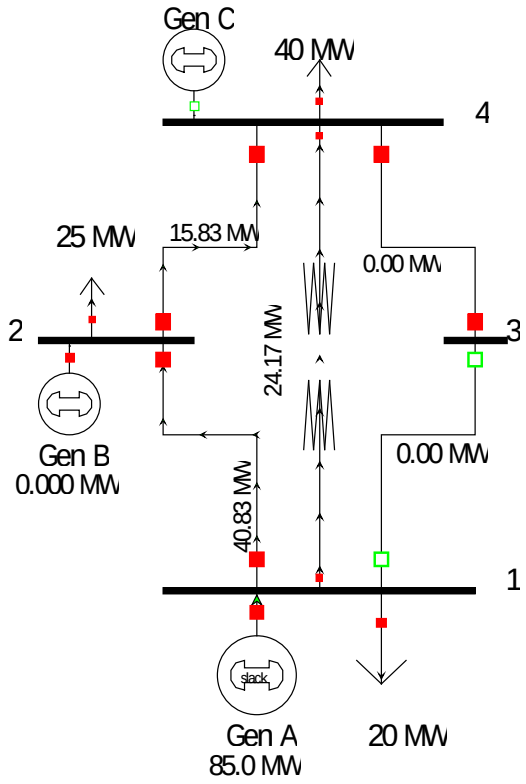
By setting the QB at 1.86 phase shift degrees, the flows at various operating conditions can be analysed.

The most severe outage (Line 1-2 outage):

The flows are shown in the diagram below; only generator A is used as Gen B is no longer needed because the congestion can be relieved by the QB.



Line 1-3 outage:



It can be concluded that by installing QB at line 1-4 and setting the QB at 1.86 phase-shift degrees, the system can operate at lower cost and N-1 secure.

Question 2

a) Explain the principles of centralised and price-based scheduling of flexible demand appliances. What are the advantages and disadvantages of each of these approaches?

[4]

Under centralised scheduling, the smart control systems of flexible demand appliances need to communicate their technical and economic characteristics to a central system entity which makes decisions on the scheduling of the appliances based on the solution of a global optimisation problem and communicates according dispatch signals to the appliances. Under a significant penetration of flexible demand appliances however, the communication and computational scalability of centralised mechanisms is at least questionable in both technical and economic terms. Transmission of the complex and diverse operational constraints and physical parameters of a very large number of demand appliances to the central entity will yield information collection and communication problems, while the vast number of decision variables and constraints associated with these appliances will create a massive computational burden to the central entity. Furthermore, centralised scheduling is likely to raise privacy concerns by the consumers, as they are not generally willing to disclose sensitive information (such as habits, preferences and appliances' properties) and be directly controlled by an external entity.

Under price-based scheduling, these scalability and privacy limitations are avoided. Without communicating their individual characteristics to the central entity, flexible demand appliances are exposed to time-differentiated prices and self-schedule their demand to minimise the electricity cost incurred by their operation. More specifically, the posted prices are higher during peak demand periods and lower during off-peak demand periods to incentivise appliances to redistribute their energy requirements towards the latter and therefore improve the efficiency of system operation. However, naive application of dynamic pricing in combination with the envisaged automation in appliances' control leads to serious loss of diversity and demand response concentration phenomena. Specifically, appliances' demand is concentrated at the periods with the lowest prices, yielding significant new demand peaks and thus inefficient system operation.

b) A simple power system, operating at a two-hour horizon, includes:

- a generator producing power s_t (MW) at hour t with a cost function $C_t(s_t) = 100s_t^2$ (£) and a maximum output limit $s^{max} = 10$MW

- inflexible demand appliances, consuming power $D_1 = 3$MW (at hour 1) and $D_2 = 2$MW (at hour 2)

- 1000 identical flexible demand appliances with a shiftable cycle of duration $T^{dur} = 1$ hour, and demand $P^{cyc} = 5kW$. These appliances are activated at hour 1 and their cycle can be delayed by $\delta = 1$ hour.

i) Under centralised scheduling, what is the optimal schedule of the flexible demand appliances at the two hours? What is the total generation cost?

[2]

ii) Assuming that price-based scheduling is employed, what should be the relation between the prices at hour 1 and 2 (denoted by λ_1 and λ_2 respectively)? What is

the resulting schedule of the flexible demand appliances at the two hours? What is the total generation cost? Is it higher or lower than the total generation cost calculated in i)? Why it is higher or lower?

[4]

Based on the data, the cycle of each flexible demand appliance can be scheduled either at hour 1 or hour 2. We denote the number of flexible demand appliances scheduled at hours 1 and 2 as N_1 and N_2, respectively.

i) Total generation must be equal to total demand at each hour, implying:

$s_1 = D_1 + N_1 * P^{cyc}$ and $s_2 = D_2 + N_2 * P^{cyc}$

Due to the quadratic form of the generator's cost function, the total generation costs are minimised when the generation profile is as flat as possible, or equivalently when:

$|s_1 - s_2|$ is minimised

Combining the above equations means that the objective is:

Minimise $|(D_1 + N_1 * P^{cyc}) - (D_2 + N_2 * P^{cyc})|$

While the operating constraints of the generator and the flexible demand appliances are satisfied:

Generator: $D_1 + N_1 * P^{cyc} \leq s^{max}$ and $D_2 + N_2 * P^{cyc} \leq s^{max}$

Flexible demand appliances: $N_1 + N_2 = 1000$

This is achieved when $N_1 = 400$ and $N_2 = 600$, i.e. when the cycle of 400 flexible demand appliances is scheduled at hour 1 and the cycle of 600 flexible demand appliances is scheduled at hour 2.

The total generation cost is $100(D_1 + N_1 * P^{cyc})^2 + 100(D_2 + N_2 * P^{cyc})^2 = £5000$

ii) The price at hour 2 should be lower ($\lambda_1 > \lambda_2$) in order to incentivise flexible demand appliances to move their demand to the off peak-hour 2 and flatten the system demand profile.

Each flexible demand appliance aims at minimising its electricity cost. Therefore, it will schedule its cycle at the hour with the lowest price. As a result:

$N_1 = 0$ and $N_2 = 1000$

The system operator schedules the generator to meet the resulting total demand:

$s_1 = D_1 + N_1 * P^{cyc} = 3MW$ and $s_2 = D_2 + N_2 * P^{cyc} = 7MW$

The total generation cost is $100(D_1 + N_1 * P^{cyc})^2 + 100(D_2 + N_2 * P^{cyc})^2 = £5800$

The total generation costs are significantly higher than the costs under centralised scheduling. This happens because of the concentration of flexible demand response at the hour with the lowest price.

c) State the main big data challenges and the primary sources of big data in power systems.

[3]

Answers should state that big data challenges include high-volume, high-velocity and high-variety information made available through increased instrumentation at all system levels.

The primary sources of big data in power systems are:

• Phasor Measurement Units (PMUs)

• Asset monitoring data from various equipment e.g. transformers

• Smart meters

• Electricity market data

d) Give three examples of big data analysis techniques. Give an example technique with an application in energy systems for supervised learning and unsupervised learning, respectively. For a transmission system operator, identify a specific task, describe how it was carried out in a traditional way and how it can be enhanced using big data.

[4]

Big data analysis techniques include:

1) Statistical models
2) Machine learning
3) Massive parallelism
4) Data visualisation
5) Data storage

….

Unsupervised learning:

Techniques: clustering, anomaly detection

Applications: Residential electricity consumption characterisation, Electricity theft detection, Big data compression

Supervised learning:

Techniques: classification, regression

Application: system security assessment, load and renewable energy forecasting, customer responsiveness quantification to time-varying prices.

In the case of the Transmission System Operator, one task that can be improved using big data analytics is that of condition awareness (can also be referred to as security assessment/dynamic analysis etc.). Traditionally, security assessment has been carried out at a limited, ad-hoc basis, largely based on empirical rules and conservative estimates. Big data analytics can be applied to data gathered from Phasor Measurement Units (PMUs) to carry out large-scale security assessment and improve situational awareness. In particular, historical PMU data also contain information on what operating points the operator should anticipate in the future. Simulations of faults can be carried out off-line to identify stability characteristics of individual operating points. Machine learning techniques (e.g. Decision Trees)

are then used to produce security rules delineating the system's region of stable operation.

e) Give a brief definition for concept of Demand Response (DR). What are the benefits/targets of DR? Give the definition of baseline estimation and illustrate the impacts of inaccurate baseline estimation. What are the current baseline estimation approaches?

[3]

Demand response (DR) is defined as change in demand as a result of a price signal, relative to the demand that would have been had the price signal not been sent.

DR is one of the most flexible and cost-effective solutions for reducing system operation costs and can displace or defer network reinforcement costs by reducing the peak demand and peak-valley difference.

Baseline estimation is the fundamental but challenging component of DR. It is performed to estimate the electricity consumption under normal conditions during the DR events. Inaccurate baseline estimation results in unfairness between the DR participants and aggregators: an overestimated baseline demand may cause aggregators to offer more compensation to customers; underestimation will influence customer responsiveness due to the insufficient rewards. Current baseline estimation approaches include: 1) average methods/similar day methods, 2) regression methods; 3) the CONTROL group methods.

Question 3

a) Discuss the concept of risk and its use in reliability analysis. Provide the definition of a risk metric used to quantify the reliability of distribution networks and one of transmission networks.

Explain why different metrics have been adopted to conduct reliability analysis for the distribution and transmission network.

[4]

The concept of risk is used to evaluate the future performance of a system. In particular, the analysis of future reliability is performed introducing risk measures. Risk measures are based on probabilistic descriptions and performance indicators.

Kolmogorov's axioms of probability allow to use the concepts of a random state and scenario S and evaluate an indicator called also impact function Im(S) on a random state selected accordingly to a probability distribution. The most common risk measure is defined as the expectation of the random variable Im≡Im(S) as follows

$$\text{Risk} = E[\text{Im}]$$

The term risk has usually a negative interpretation since it is seen as a measure of the absence of reliability, however the same concept can also be used for positive indicators.

The expectation of the impact is the simplest definition of risk and it is referred as risk neutral.

There are more general risk definitions such as risk-averse approaches that focus on the tail of distributions, or the general class of coherent risk measures.

A risk measure returns a single value or a set of numbers in case of multiple experiments.

The distribution of an indicator is referred as a risk profile of that indicator

Commonly more than one performance indicator is monitored at the same time because different indicators quantify different aspects of reliability.

Any of the following distribution risk metrics can be chosen:

- Expected Customer Minutes Lost (E[CML])

- Expected Customer interruptions (E[CI])

Any of the following transmission risk metrics can be chosen:

- EENS / EUE / EEU Expected Energy Not Supplied / Expected Unserved Energy / Expected Energy Unserved: Expected amount of energy that is not supplied in a given period (usually MWh/year).

- LOLP Loss of load probability: Probability of inability to supply all customer loads (at a random moment in time).

- LOLE Loss of load expectation: Expected number of periods in which the system is unable to supply all load (usually expressed in hours/year –sometimes days/year)

- LOLF Loss of load frequency: Average rate of occurrence of 'loss-of-load events' (usually occurrences/year).

Distribution network reliability analysis is often performed separately from analysis of higher voltage levels for multiple reasons

- Transmission and distribution systems operate on different scales and geographical areas.
- They are owned and operated by different companies
- Smart Grid technologies have profound effects on reliability at the distribution level
- Distribution network analysis is more customer-centric (or load point centric) than system centric.
- The bulk of customer interruptions originates in the distribution system. Failures in distribution system components usually do not affect the bulk system, but they occur relatively often.

MARKING:

- 2 point: 1 point to correctly give a commonly used definition of risk for reliability indicators. 1 additional point for a more comprehensive discussion on the concept of risk and its use in reliability analysis.
- 1 point: correctly identifying E[CML] or E[CI] as distribution risk metrics and EENS / EUE / EEU or LOLP or LOLE or LOLF as transmission risk metrics.
- 1 point: providing an acceptable explanation why different metrics for the distribution and transmission network have been adopted.

b) Consider a section of a distribution network serving 12000 customers. The outage events occurred in one year are reported in the table below.

Customers Affected	**Duration**
3500	20 minutes
700	20 hours
8000	1 minute
1100	45 minutes
5000	3 minutes
10	1 day 3 hours
900	30 minutes

Table 1- Outage events

Compute CML and CI for this section of the network. Discuss the difference between CML and expected CML.

[3]

Due to the 3-minute minimum requirement for CML and CI, the recorded event that affected 8000 customers for 1 minutes will be excluded from CML/CI calculations.

$$CML=\frac{\sum_{events} N_{affected} T_{minutes}}{N_{customers}} = \frac{3500*20+700*20*60+5000*3+1100*45+10*27*60+900*30}{12000} = 84.81 \text{ minutes}$$

$$CI=100\times\frac{\sum_{events} N_{affected\ customers}}{N_{customers}} = 100\times\frac{3500+700+5000+1100+10+900}{12000} = 93.42$$

'CML' refers to the realised performance in a past year and 'Expected CML' to the average performance in a future year.

MARKING:
- 1 point: correctly excluding the 1-minute event

- 1 point: providing correct expressions for CML and CI and correct distinction between CML and E[CML]

- 1 point: correct numbers

c) Consider the distribution network in Figure 2. The number of customers at each load point (a, b, c, d) is reported in Figure 2. The customers at load points a and b are of residential type. Each customer at load points a and b has an average power consumption of 0.4kW and a Value of Lost Load (i.e. interruption cost per unit of lost energy) equal to 5000£/MWh. The customers at load point c are commercial, while at load point d are of industrial type. Each customer at load point c has an average power consumption of 2kW and at load point d of 4kW. The Value of Lost Load at load point c is 20000£/MWh at load point d is 28000£/MWh.

The availabilities of the transformers (T1, T2) and the feeders (F1, F2, F3, F4) are reported in Figure 2. The availability of the Bus c-d is 99.95%. Assume that all components failures are independent and they all last more than three minutes. The individual capacity of T1 and T2 are always sufficient to serve all customers at load points a, b c and d. The network is radially operated but the normally open points (NOPs) can ensure some redundancy. They can be closed to resupply load points in case of faults in the network.

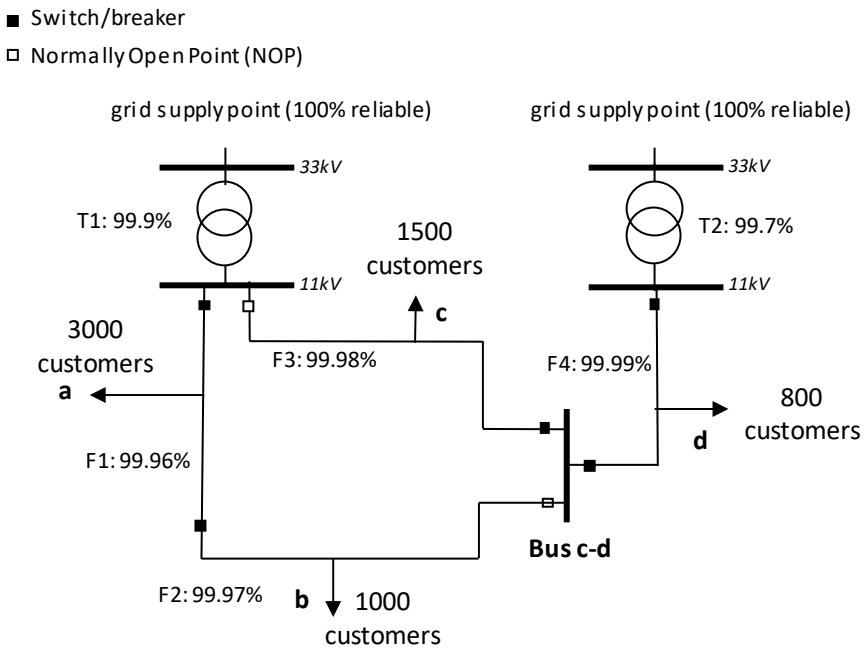


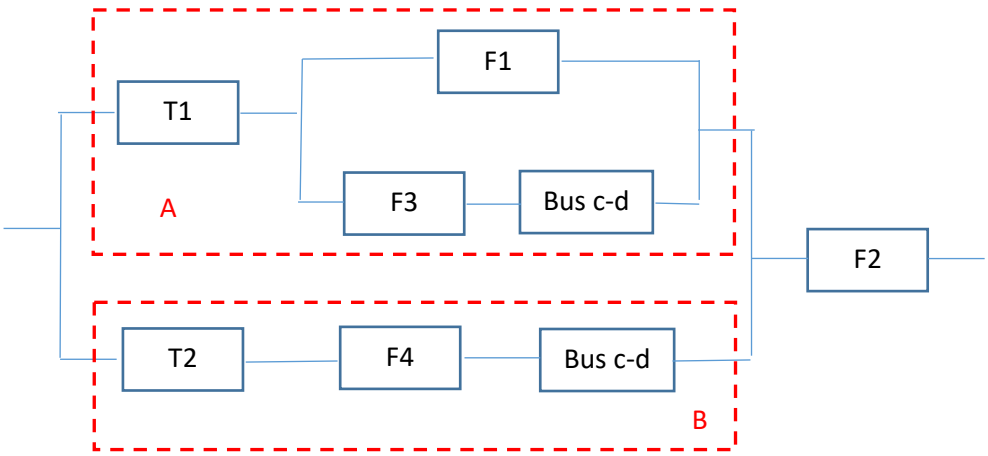
Figure 2- Network diagram

(1) Calculate the probability that each load point is connected to a grid supply point (one expression for each load point). Note that the failure of a transformer implies that the transformer bus will fail too.

Let p(x) denote the availability of component x and a_x the probability that customers at load point x are connected to a grid supply point. The probabilities of each component being connected to a grid supply point are the following:

$$a_a = (1 - (1 - p(T1))(1 - p(T2) \cdot p(Bus_{c-d}) \cdot p(F2) \cdot p(F4))) \cdot p(F1)$$
$$= (1 - (1 - 0.9990) \cdot (1 - 0.997 \cdot 0.9995 \cdot 0.9997 \cdot 0.9999))$$
$$\cdot 0.9996 = 0.9996$$

Load b can be served using three different alternatives as follows



The events A and B are not independent and consequently $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ where

$$P(A) = p(T1) \cdot \Big(1 - \big(1 - p(F1)\big) \cdot (1 - p(F3) \cdot \mathrm{p}(\mathrm{Bus_{c-d}})\ )\Big) = 0.9990$$

$$P(B) = \mathrm{p(T2)} \cdot \mathrm{p(F4)}\ \cdot \mathrm{p}(\mathrm{Bus_{c-d}}) = 0.9964$$

and $B = T2 \cap F4 \cap \ \mathrm{Bus_{c-d}}$, $A = T1 \cap C$ with $C = F1 \cup (F3 \cap \mathrm{Bus_{c-d}}) = (F1 \cup F3) \cap (F1 \cup \mathrm{Bus_{c-d}})$

Consequently

$$P(A \cap B) = P(T1 \cap C \cap T2 \cap F4 \cap \mathrm{Bus_{c-d}})$$
$$= p(T1)\, p(T2) p(F4) p(C \cap \mathrm{Bus_{c-d}}) =$$
$$= p(T1)\, p(T2) p(F4) p\big((F1 \cup F3) \cap (F1 \cup \mathrm{Bus_{c-d}}) \cap \mathrm{Bus_{c-d}}\big) =$$
$$= p(T1)\, p(T2) p(F4) p\big((F1 \cup F3) \cap \mathrm{Bus_{c-d}}\big) =$$
$$p(T1)\, p(T2) p(F4) p(\mathrm{Bus_{c-d}})(1 - (1 - p(F1))(1 - p(F(3)) = 0.9954$$

$$\mathrm{a_b} = (\mathrm{P(A)} + \mathrm{P(B)} - \mathrm{P(A \cap B)}) \cdot \mathrm{p(F2)} = 0.9997$$

$$\mathrm{a_c} = (1 - \big(1 - \mathrm{p(T1)}\big)\big(1 - \mathrm{p(T2)} \cdot \mathrm{p}(\mathrm{Bus_{c-d}}) \cdot \mathrm{p(F4)})) \cdot \mathrm{p(F3)}$$
$$= \big(1 - (1 - 0.9990) \cdot (1 - 0.997 \cdot 0.9995 \cdot 0.9999)\big) \cdot 0.9998$$
$$= 0.9998$$

$$\mathrm{a_d} = \Big(1 - \big(1 - \mathrm{p(T2)}\big)(1 - \mathrm{p(T1)} \cdot \mathrm{p(D)}\ \cdot \mathrm{p}(\mathrm{Bus_{c-d}})\ )\Big) \cdot \mathrm{p(F4)}$$
$$= \big(1 - (1 - 0.9970) \cdot (1 - 0.9990 \cdot \mathrm{p(D)} \cdot 0.9995)\big) \cdot 0.9999$$
$$= 0.9999$$

where $\mathrm{p(D)} = 1 - \big(1 - \mathrm{p(F3)}\big)\big(1 - \mathrm{p(F1)}\ \cdot \mathrm{p(F2)}\big) = 1 - (1 - 0.9998)(1 - 0.9996 \cdot 0.9997)$

MARKING:

- 4 points: correct expressions for $\mathrm{a_a}$, $\mathrm{a_b}$, $\mathrm{a_c}$ and $\mathrm{a_d}$ (1 point for each load point)
- 2 points: correct numbers

(2) Compute the CML for the given network and the expected annual cost of interruptions for this network.

[4]

$$\mathrm{CML} = 8760\mathrm{h} \cdot \frac{60\mathrm{min}}{\mathrm{h}} \cdot \frac{\sum_{\mathrm{loadpoints}} \mathrm{N_{customers\ at\ loadpoint}}\ \cdot\ \mathrm{unavailability_{loadpoint}}}{\mathrm{N_{customers}}}$$

$$= 8760\mathrm{h} \cdot \frac{60\mathrm{min}}{\mathrm{h}} \cdot \frac{3000 \cdot (1 - \mathrm{a_a}) + 1000 \cdot (1 - \mathrm{a_b}) + 1500 \cdot (1 - \mathrm{a_c}) + 800 \cdot (1 - \mathrm{a_d})}{6300}$$

$$= 158.89\ \mathrm{min}$$

The expected customer interruption cost for an industrial customer at load point X (either a or b) is given by:

$$\mathrm{E}\left[\mathrm{cost}_{\mathrm{ind,X}}\right] = (1-\mathrm{a_X}) \cdot 0.4\mathrm{KW} \cdot 8760\mathrm{h} \cdot \frac{\frac{5000£}{\mathrm{MWh}}}{\frac{1000\mathrm{KWh}}{\mathrm{MWh}}} = (1-\mathrm{a_X}) \cdot £17{,}520$$

The expected customer interruption cost for a residential customer at load point c is given by:

$$\mathrm{E}\left[\mathrm{cost}_{\mathrm{res,c}}\right] = (1-\mathrm{a_c}) \cdot 2\mathrm{KW} \cdot 8760\mathrm{h} \cdot \frac{\frac{20000£}{\mathrm{MWh}}}{\frac{1000\mathrm{KWh}}{\mathrm{MWh}}} = (1-\mathrm{a_c}) \cdot £350{,}400$$

The expected customer interruption cost for a residential customer at load point d is given by:

$$\mathrm{E}\left[\mathrm{cost}_{\mathrm{res,d}}\right] = (1-\mathrm{a_d}) \cdot 4\mathrm{KW} \cdot 8760\mathrm{h} \cdot \frac{\frac{28000£}{\mathrm{MWh}}}{\frac{1000\mathrm{KWh}}{\mathrm{MWh}}} = (1-\mathrm{a_d}) \cdot £981{,}120$$

The total expected cost corresponds to:

$$\mathrm{E}[\mathrm{cost}] = £\,17{,}520 \cdot [(1-\mathrm{a_a}) \cdot 3000 + (1-\mathrm{a_b}) \cdot 1000] + £350{,}400 \cdot [(1-\mathrm{a_c}) \cdot 1500] + £981{,}120 \cdot [(1-\mathrm{a_d}) \cdot 800] = £215{,}583$$

MARKING:
- 1 point: correct expression for CML
- 2 points: correct expression for total expected cost
- 1 point: correct numbers

d) The Weibull distribution (Weibull 1951) is a generalization of the exponential distribution and is commonly used to represent component lifetimes in various types of engineering applications. The two-parameters probability density function is

$$f(t) = \lambda\beta\,(\lambda t)^{\beta-1}\, e^{-(\lambda t)^{\beta}}$$

where β>0 is a shape factor and λ>0 a scale factor

(1) Compute the MTTF expression for a failure process modelled with the Weibull distribution. Hint: Remind that the gamma integral is defined as $\Gamma(k) = \int_0^{\infty} y^{k-1}e^{-y}dy$ and given k the value of the integral is tabulated. Given $\lambda = 0.01\ h^{-1}$ and the provided table draw the MTTF for a few values of β.

(2) Is the Weibull distribution memoryless?

(Hint: A distribution satisfies the memoryless property if $P\{T > t\} = P\{T > t + s \mid T > s\}$ for $t > 0$, $s > 0$, i.e the conditional probability does not depend on s.

(3) Compute analytically the hazard rate for the Weibull distribution and draw it for different values of the parameters.

(Hint: $\lambda(t)dt$ is the probability that a device of age t will fail in the small interval of time t to $t + dt$).

[3]

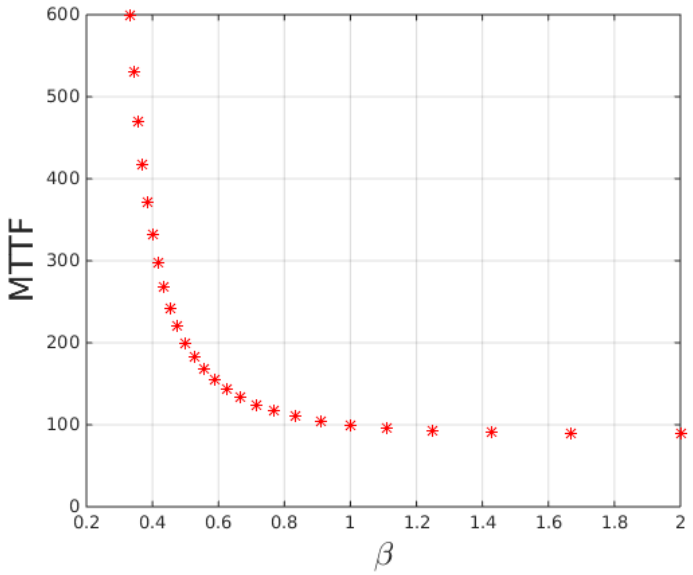
(1) The MTTF is, by definition, the mean value of the stochastic variable and for a Weibull distribution is

$$E[t] = \int_0^{\infty} t\,\lambda\,\beta\,(\lambda\,t)^{\beta-1}\,e^{-(\lambda\,t)^{\beta}}dt$$

Applying the change of variable $u = (\lambda t)^{\beta}$ we obtain

$$E[t] = \frac{1}{\lambda}\int_0^{\infty} u^{\frac{1}{\beta}}\,e^{-u}du = \frac{1}{\lambda}\,\Gamma\left(1+\frac{1}{\beta}\right) = \frac{1}{\lambda\beta}\,\Gamma\left(\frac{1}{\beta}\right)$$

where $\Gamma(\cdot)$ is the gamma function and the last is obtained using the recursion $\Gamma(k+1) = k\,\Gamma(k)$. Note that if $\frac{1}{\beta}$ is an integer, $\Gamma\left(\frac{1}{\beta}\right) = \left(\frac{1}{\beta}-1\right)!$ holds.



(2) The memoryless property can be investigated writing P{T>t+s | T>s} for the Weibull distribution.

$$P\{T > t+s \mid T > s\} = \frac{P\{T > t+s \text{ and } T > s\}}{P\{T > s\}} = \frac{P\{T > t+s\}}{P\{T > s\}} = \frac{e^{-(\lambda(t+s)^{\beta})}}{e^{-(\lambda s)^{\beta}}}$$

From the last expression we observe that the dependence on s can be only removed when the positive shape parameter β is equal 1.

(3) Denote as F(t) the cumulative distribution function. By definition

$$h(t)dt = P\{t < T \leq t + dt \mid T > t\} = \frac{P\{t < T \leq t + dt\}}{P\{T > t\}}$$

Then

$$h(t) = \lim_{dt \to 0} \frac{F(t + dt) - F(t)}{dt} \frac{1}{P\{T > t\}} = \frac{f(t)}{1 - F(t)}$$

For the Weibull distribution the hazard rate is

$$h(t) = \lambda \beta (\lambda t)^{\beta - 1}$$

MARKING:
- 1 point: correctly giving the definition and expression of MTTF
- 1 point: correctly showing that the Weibull distribution is not memoryless
- 1 point: correctly giving the expression of the hazard rate for the Weibull distribution

Question 4

a) There are two gas-fired power generating units. One (Generator 1) has failed 20 times in the past 8 years and repairs have taken 1 week on average, while another (Generator 2) has failed 10 times in the past 8 years and repairs have taken 2 weeks on average. Assuming that these are representative of their overall reliability, what is the mean time to failure (MTTF)? What is the availability of each unit? Compare and comment on the performance of the two units.

[4]

Assumption: The 8 years represents a whole number of failure-repair cycles

(i.e. representative).

For generator 1, the mean time between failures (MTBF) is 8 yeas/20 = 0.4 years = 20.86 weeks. The MTTR is 1 week. Therefore, MTTF=MTBF – MTTR = 19.86 weeks. The availability is av=MTTF/MTBF = 0.95

For generator 2, the mean time between failures (MTBF) is 8 yeas/10 = 0.8 years = 41.72 weeks. The MTTR is 1 week. Therefore, MTTF=MTBF – MTTR = 39.72 weeks. The availability is av=MTTF/MTBF = 0.95

[3]

The two generators have the same availability rate for long term operation, but the frequency of failure is higher for Generator 1 while less time is required for repair. For short-term operation (daily), Generator 1 has more adversary impact than Generator 2.

[1]

b) Consider a large-scale power station that has three operational states: ‘failed’, ‘fully operational’ and ‘partially operational’. A Markov model is used to model transitions between states. The transition rates are as follows:

(1) from any operational state, complete failures occur at a rate of 1 per year;

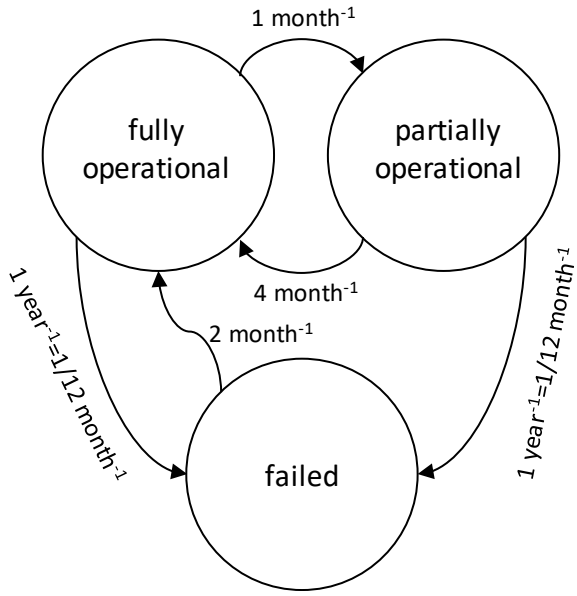
(2) repairs from the failed state are carried out at a rate of 2 per month and always result in a fully operational unit;

(3) partially operational units return to full operation at a rate of 1 per month:

(4) fully operational units enter a partially operational state at a rate of 4 per month;

i) Draw the corresponding continuous-time Markov model.

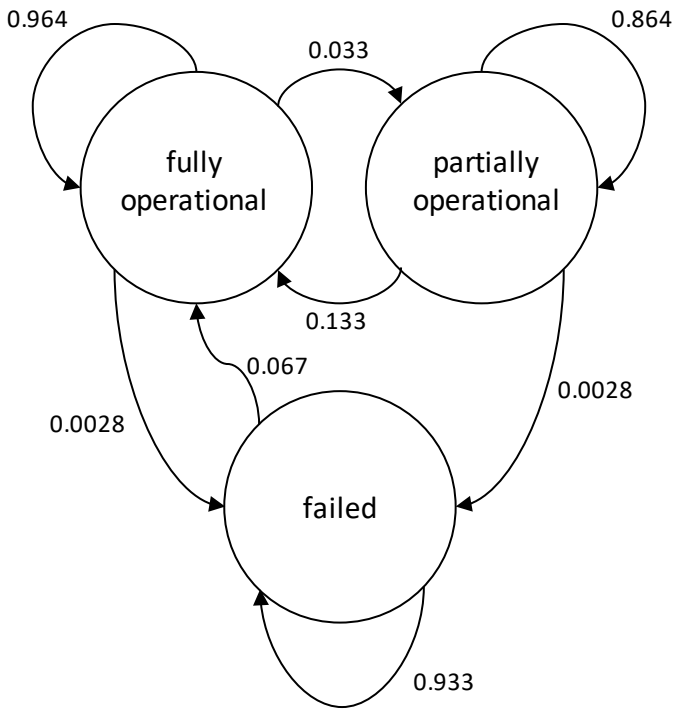
[2]



[2]

ii) Draw a discrete time Markov chain for this process, assuming a time step of 1 day. Include all non-zero transition probabilities.

[2]



[2]

c) A system operator is designing a forced islanding scheme: in case of an acute blackout threat, the scheme preventively splits the power system into separate electrical islands. Such a scheme must have a high degree of dependability, but also be resilient against spurious activations.

The proposed design is as follows. The state of the power system is monitored at three locations, each equipped with a monitoring station. If a monitoring station detects a critical system condition it sends an alert to the control room over a dedicated communication channel. The decision to island the system is taken by a computer in the control room when it receives alert signals from at least two sites. To provide redundancy, there are three identical computer systems that receive identical signals and that can each activate the islanding signal.

Assuming that the critical condition is observable at each of the three monitoring stations (if they are operational), independence between components and an availability of 95% for monitoring stations, 98% for communication channels and 90% for the computer, quantify the probability that a critical operating condition is successfully detected.

[5]

First, compute the availability of a monitoring-communication pair:

Av(mici)=0.95*0.98=0.931 [1]

2-out-of-3 voting is 3*0.931^2*(1-0.931) + 0.931^3=0.1794+0.8069=0.9863 [2]

The availability of computer system is 1-(1-0.9) ^3=0.999 [1]

The combined availability is 0.999*0.9863= 0.9853 [1]

A snapshot Monte Carlo simulation is used to estimate the expected cost of interruptions. After N=5,000 samples the best estimate of the cost is £108,762, with a standard error of £2581.

i) How many samples will be required (approximately, in total) to reduce the standard error of the mean to £1000?

[2]

The standard error after 5,000 samples is £2581. The stander errors scales as

$se=\sigma/((n)^{0.5})$, [1]

$N'=N*(2581/1000)^2=33308$ [1]

ii) How many samples will be required (approximately) to reduce the width of the 95% confidence interval for the mean to £1000?

[2]

To reduce the 95% confidence interval to £1000, we require that both tails of the interval (1.96 standard errors each) fit within the range. This is achieved when the standard error (σ') satisfies

$2*1.96*\sigma' = 1000$ [1]

Hence, $\sigma'= 255$

Using the same formula as used in (h), the required number of samples becomes

$N'=N*(2581/255)^2=512231$ [1]

d) State examples of physical threats and cyber threats to reliable power system operation. For each category, provide one example of an accidental/natural threat and one example of a malicious threat (i.e. provide four examples in total).

[3]

Examples (from lecture slides)

	Malicious	Accidental
Physical	Terrorism, Vandalism	Weather impacts, mechanical failure
Cyber	Hacking, Jamming of communication	Communication errors, Software bugs